consICA: MULTIMODAL DATA DECONVOLUTION, INTEGRATION AND ELUCIDATION OF BIOLOGICAL PROCESSES IN CANCER RESEARCH

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consICA [1]

INTRODUCTION

Analyzing cancer-related multiomics data, we need to overcome data complexity, tumor heterogeneity and technical biases which mask the important biological signals. This complexity is caused by natural variability in cell type proportions and clonal among tumor cells. Additionally, technical biases between experimental platforms may limit the direct comparison of patient data coming from different sources, especially mapping to large public datasets. We developed a Bioconductor package *consICA* to decipher molecular signals and resolve data intricacies.

APPROACH

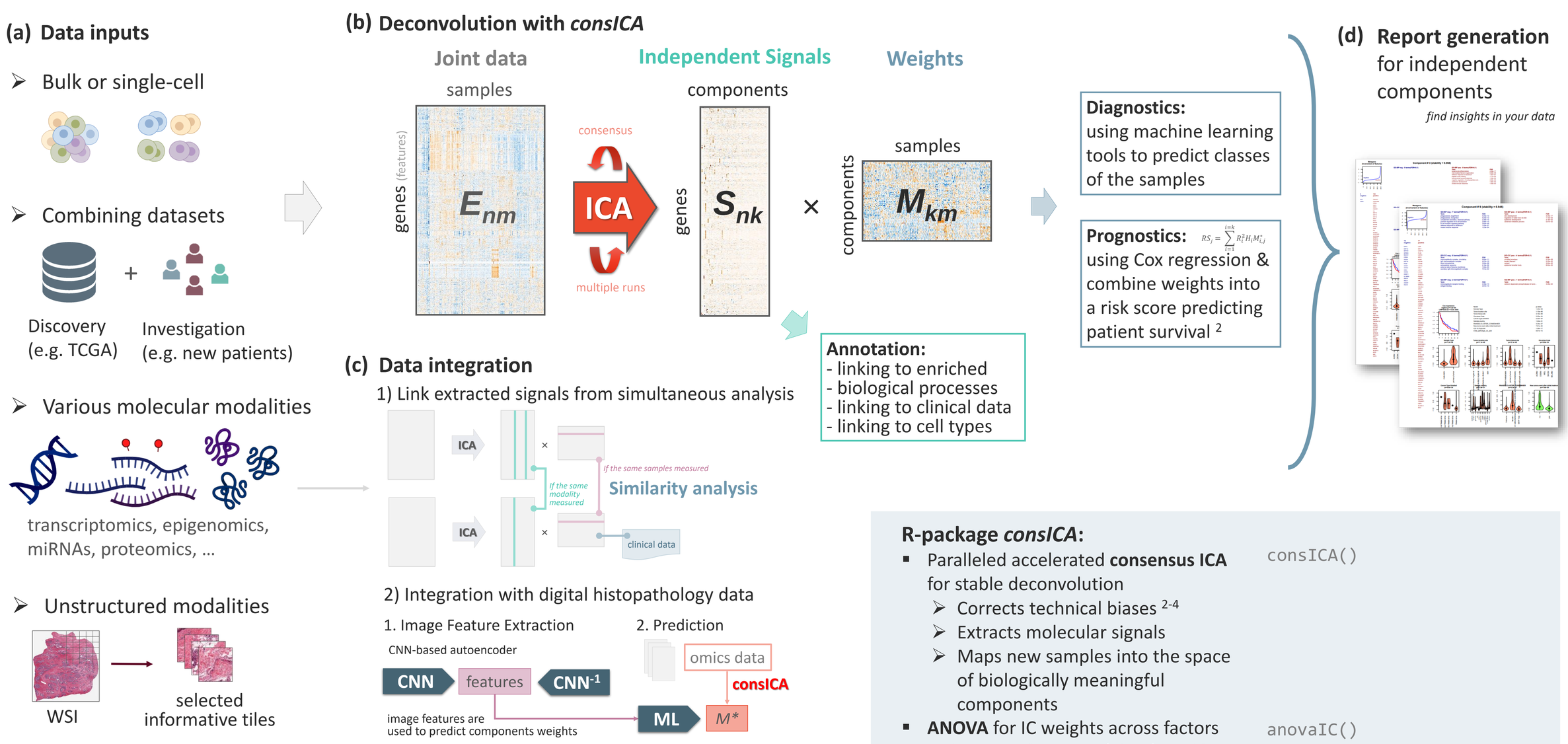


Fig 1. (a) *consICA* allows deconvolution of various modalities of data depending on the analysis goals. (b) Independent component analysis (ICA) decomposes combined matrix into meaningful signals (S) and weights (M). Independent components could be annotated with biological processes / cell subtypes, while M could be linked to patient groups and patient survival. (c) For data integration the comparative analysis could be used or integration via classical ML models with images features. (d) Automatic report accumulates all information of independent components.

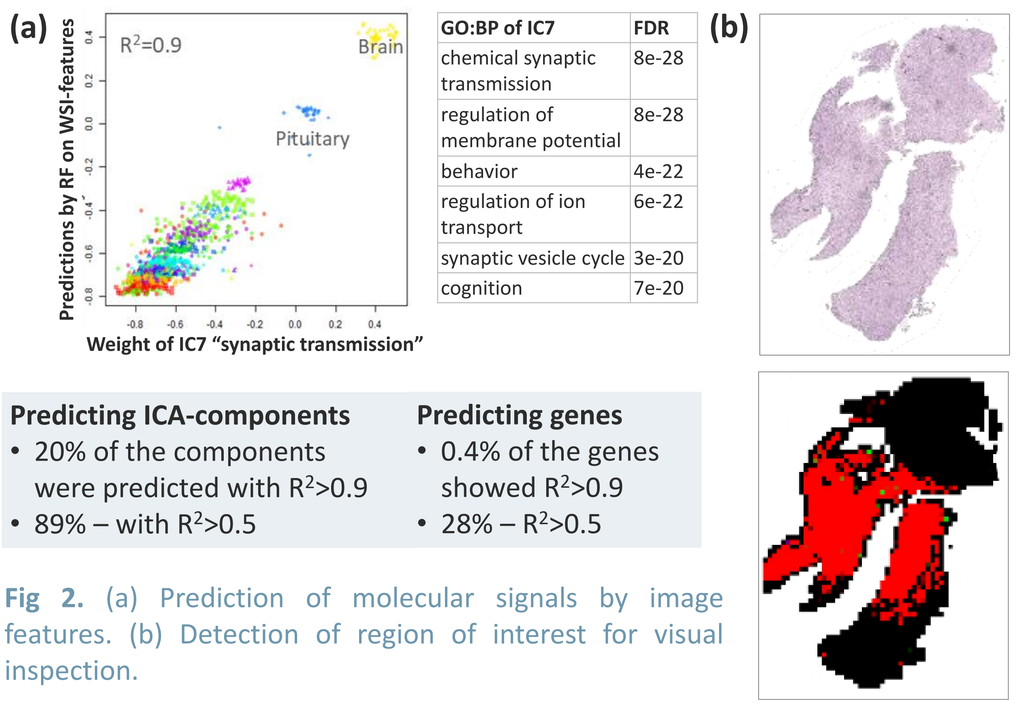
R-package *consICA*:

- Paralleled accelerated **consensus ICA** for stable deconvolution
  - Corrects technical biases [2-4]
  - Extracts molecular signals
  - Maps new samples into the space of biologically meaningful components
- **ANOVA** for IC weights across factors
- Estimate **variance** explained per IC
- Paralleled functional **enrichment analysis** with *topGO*
- Estimate **similarity** between GO lists
- **Survival** analysis
- Generate automatic **report**

```
consICA()
anovaIC()
estimateVarienceExplained() &
  plotICVarienceExplained()
getGO()
overlapGO()
survivalAnalysis()
saveReport()
```

RESULTS

Multiomics and digital histopathology data integration



GO:BP of IC7	FDR
chemical synaptic transmission	8e-28
regulation of membrane potential	8e-28
behavior	4e-22
regulation of ion transport	6e-22
synaptic vesicle cycle	3e-20
cognition	7e-20

Predicting ICA-components
- 20% of the components were predicted with R^2>0.9
- 89% – with R^2>0.5

Predicting genes
- 0.4% of the genes showed R^2>0.9
- 28% – R^2>0.5

Fig 2. (a) Prediction of molecular signals by image features. (b) Detection of region of interest for visual inspection.

Drug resistance corresponding signals in melanoma cell lines

NRASmut melanoma cell lines SKMel30, IPC298, MelJuso and primary line M20 (NRASmut patient) were treated with MEK1/2 and CDK4/6 inhibitors[5], the scRNA-seq were done at ● d00 ● d01 ● d04 ● d33 days after treatment.

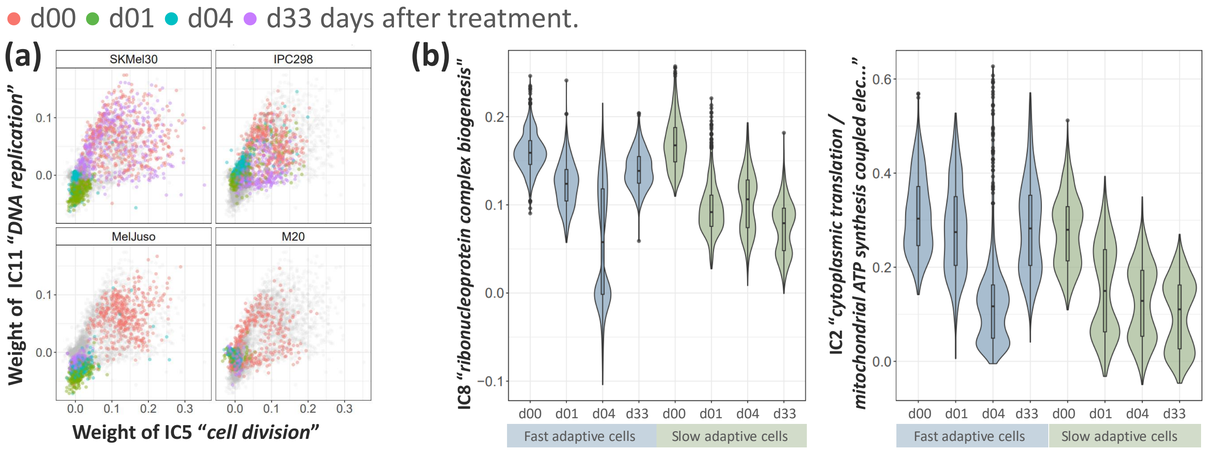


Fig 3. (a) Cells in components corresponded to cell cycle. SKMel30 and IPC298 include proliferated cells on *T33* time point in opposite to MelJuso and M20. (b) Components show differences in cells with fast and slow adaptivity: decreasing of signal on 1, 4 days after treatment but returning the signal level on 33th day for fast adaptive cells and decreasing during all period for slow adaptive cells.

References
1.https://bioconductor.org/packages/consICA
2.Nazarov et al., **2019, BMC Med Gen** 12:132
3.Sompairac et al., **2019, Int J Mol Sci** 20(18)
4.Captier et al., **2022, Bioinf** 38(10)
5.Randic et al., **2023, Cell Rep** 42(7):112696

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